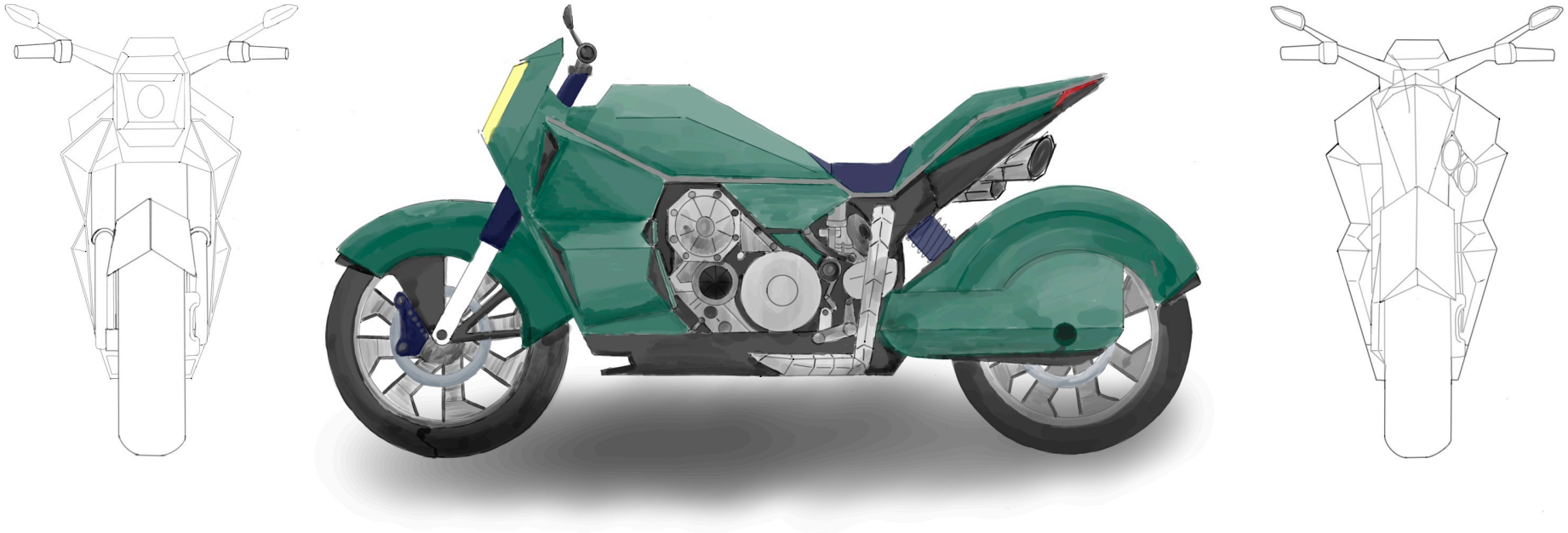
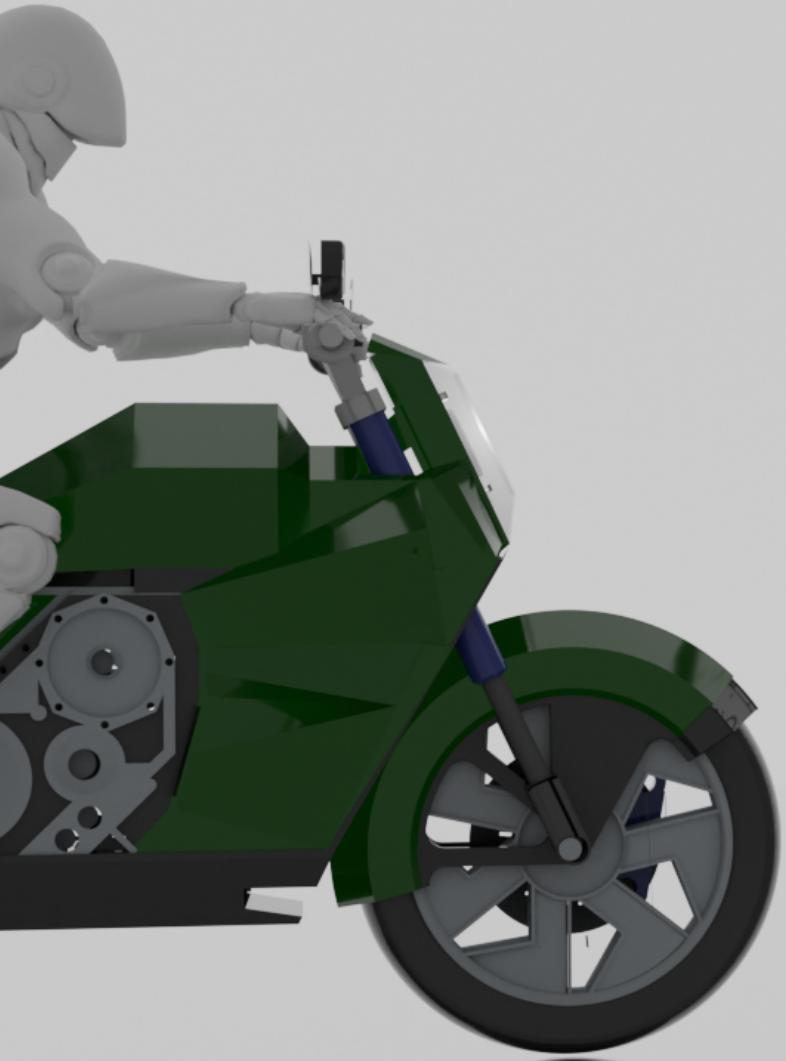
VEHICLE AERODYNAMICS

Francesco Rastelli





CONTENT

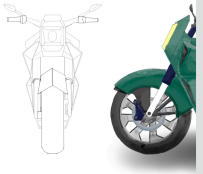
1. SUMMARY OF DESIGN PROCESS
2. INSPIRATION AND EXPERIMENTATION
3. SKETCHES
4. CAD MODEL AND RENDERS
5. DRAG ANALYSIS AND HAND CALCULATIONS
6. CFD MODEL
7. PRESSURE AND VELOCITY PLOT
8. 3D PRESSURE DISTRIBUTION
9. 3D FLOW TRAJECTORIES
10. COMPARING TO REAL WORLD EXAMPLES
11. IMPROVEMENTS

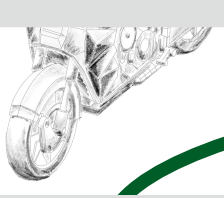
01 SUMMARY OF DESIGN PROCESS

Sketching

Initial sketches were done while looking at existing designs for inspiration which I personally like before looking at changes that could be made.

Once a **preliminary design** was produced, higher quality **sketches** were created of both side, front and back view; with only the side view being coloured after some experimentation. Different views were chosen in order to transform them into **CAD** more easily by importing them into the file.






CAD and renders

CAD work started off in **Onshape** to create the body of the bike off the sketches which were made previously. This was then transported to **Blender** so that we could add the mannequin to it and render it. However, issues arose with exporting this as we could only export it as a mesh, meaning we had to bring it into **Fusion** to transform it back into a solid body. This was then booleaned together in onshape so that **Solidworks** would accept it as a complete body without giving Non-Manifold Body errors.

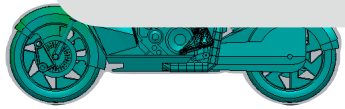




CFD analysis

CFD analysis run smoothly once solid works finally started to work.

The results from the 2D pressure and velocity **plots** were especially helpful for the analysis at the end and to work out areas which needed improvement. The 3D flow trajectories also helped with this as they showed areas of **high turbulence** effectively.

Improvements

Looking at where there were issues in my design and areas in which low and high pressure zones were forming, areas of turbulence and other CFD **results** allowed me to come up with these improvements:

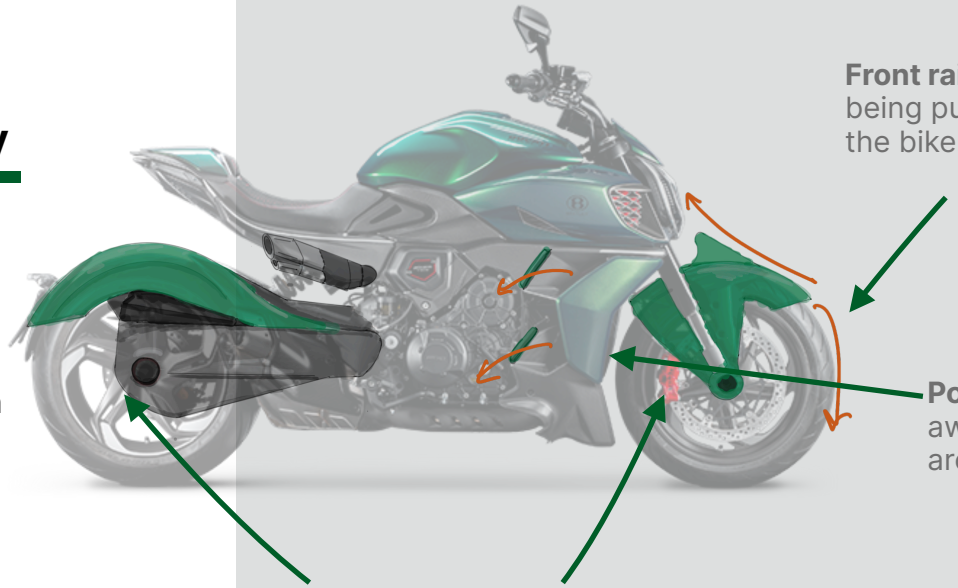
- Moving the seat back and lowering body **position**
- Increasing dimensions of **windscreen**
- Lower **angle** of attack on the front headlight
- Changing the **shape** of the back to reduce turbulence.

02 INSPIRATION AND EXPERIMENTATION

Ducati Diavel for Bentley

An existing design I took a lot of **inspiration** from is the Ducati Diavel For Bentley. On it you can see changes I thought about introducing to it for aerodynamic purpose.

Many aspects of this bike can be seen in my final design, especially with the layout of the **front intakes** and of the ones on the side.



Front raised hump leads to air being pushed over the front of the bike.

Post-intake flaps push air out away from the bike and around the user's body.

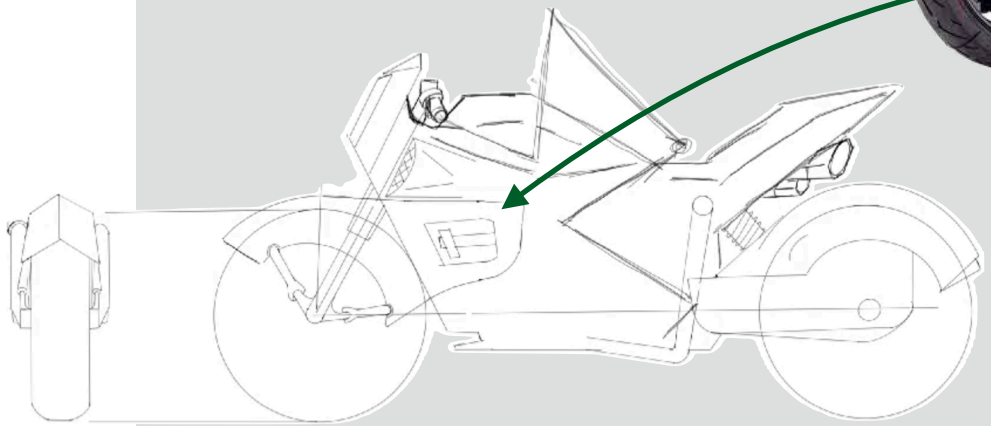
Larger wheel covers to make them more aerodynamic.



Front suspension

Experimentation occurred in the first sketches around the use of a front suspension system similar to what is found on the **Arc Vector**. However, the over complexity of this system led to me scrapping it for a simpler design.

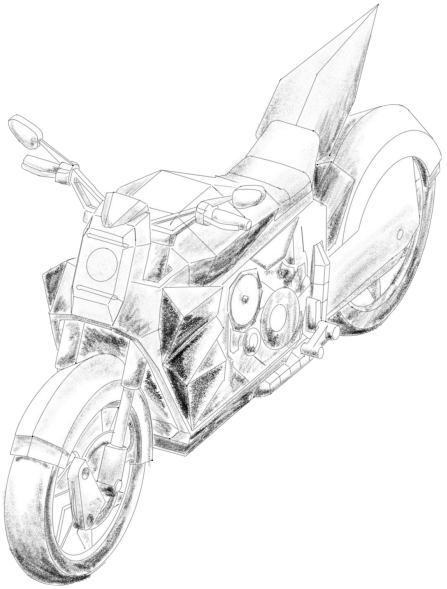
Possible benefits of this system regarding **aerodynamics** could include the ability to push air around the bike in a V shape from the front stopping it from getting trapped behind the wheel in between it and the body.



03 SKETCHES

The design

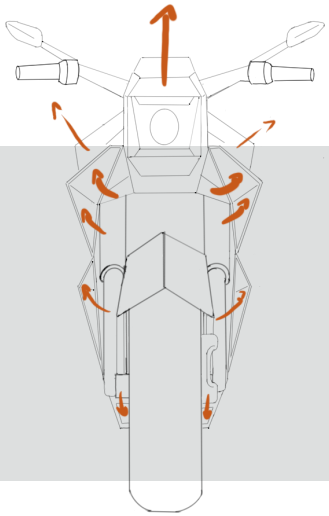
The design was developed and played around with until I came to the one you can see below. With it, I attempted to play around with both having very **rounded** shapes around the wheels, but which were then opposed by the very **angular** and aggressive body shapes. Components were then highlighted such as the forks, rear suspension, breaks and seat with a dark blue to **contrast** them against the green body.

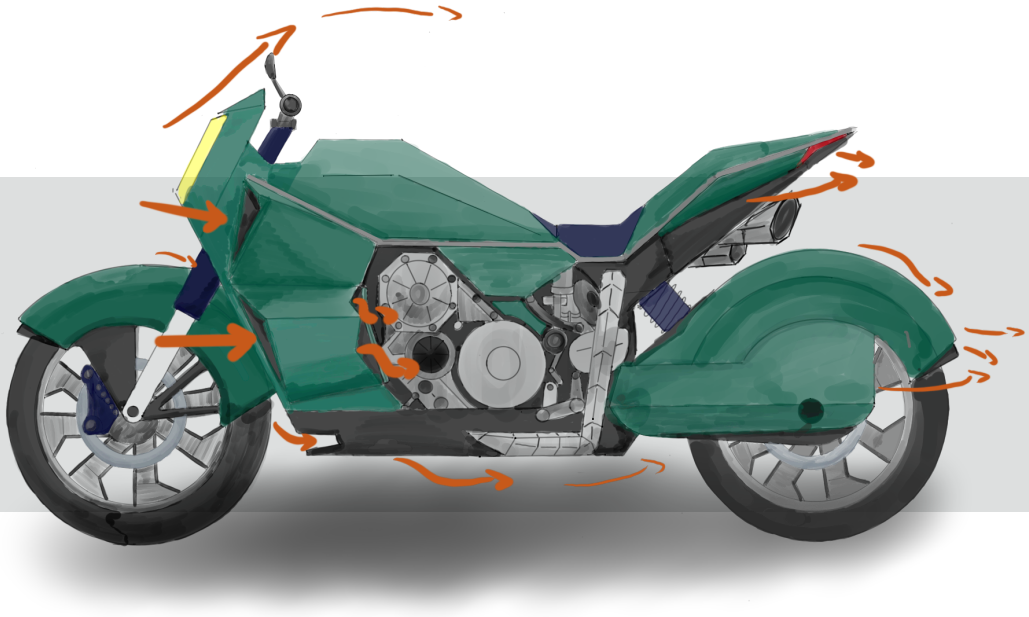


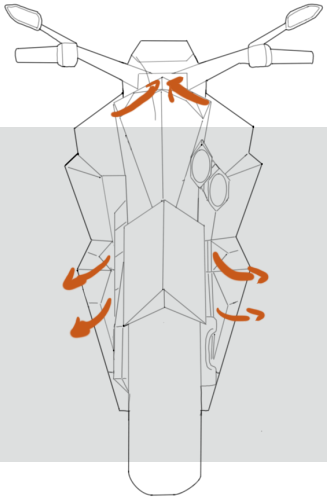
Aerodynamics thoughts:

The orange lines around the sketches attempt to portray how I envision the **airflow** to go.

The large angular light at the front is supposed to act like a **windscreen** and push air over the rider; while the intakes on the front channel air **around** them and one on the button helps funnel air that is trapped behind the wheel **underneath** the bike.







04 CAD MODEL AND RENDERS

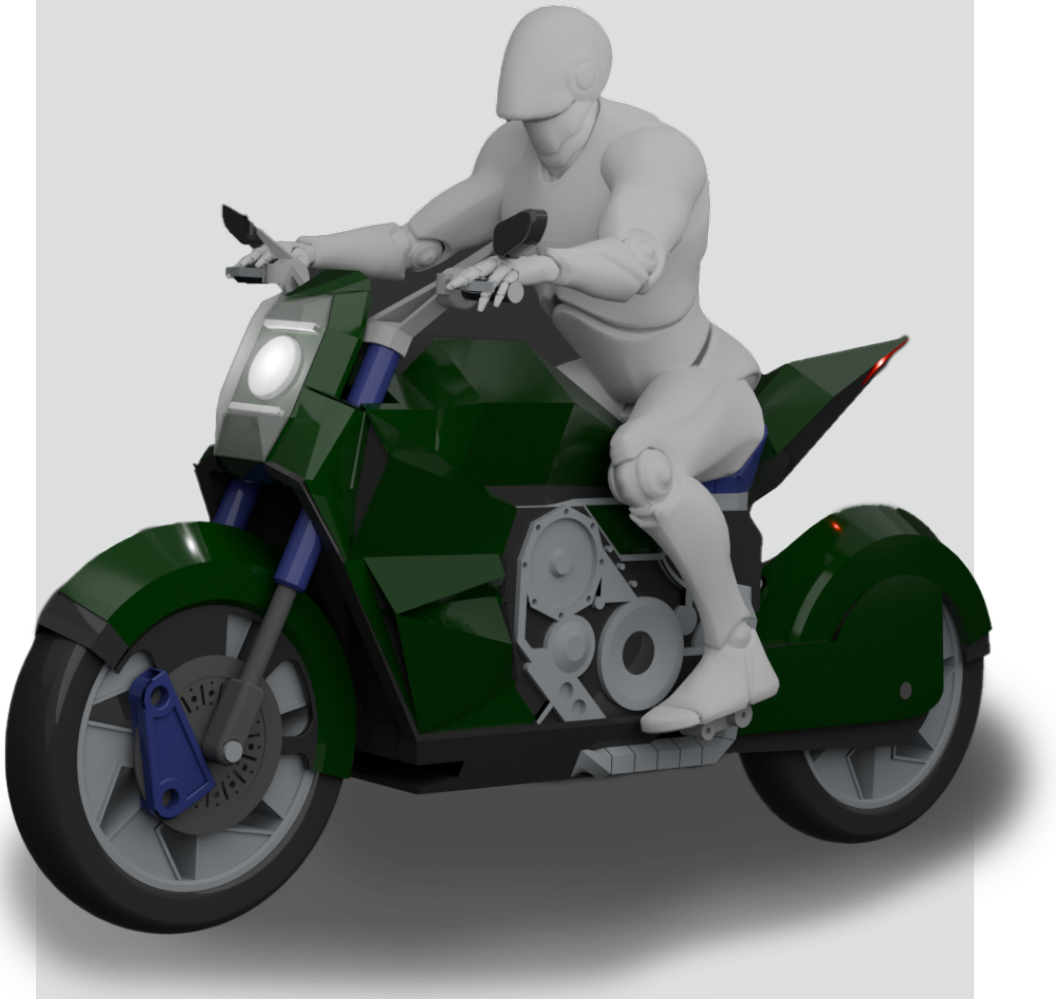
Dealing with the CAD

Transforming the design to CAD was challenging but through the use of planes, lofts and other processes I was able to make the motorbike **model**.

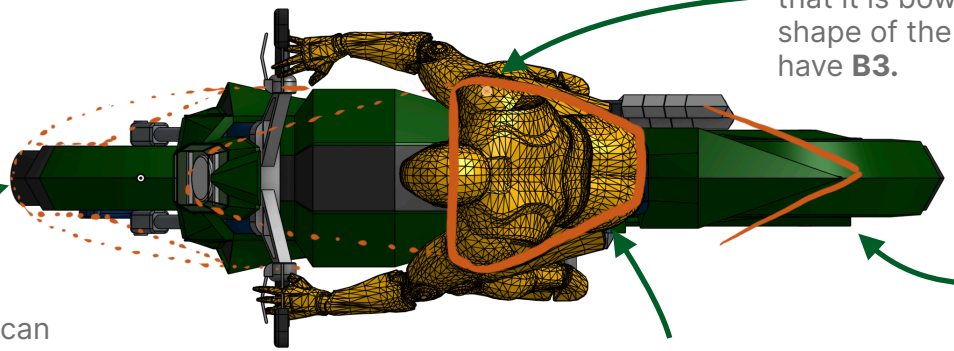
However, getting the mannequin to work was a lot more effort; it required me to learn to use Blender CAD in order to be able to **mould** it into the position in which it would be sitting on the actual motorbike. I then had to bring it back to Fusion as a .obj file and transform it into a **solid object** which would then have all 96 components booleaned together so that they became a singular part in order to allow Solidworks to accept the **geometry** and issues with it.







05 DRAG ANALYSIS AND HAND CALCULATIONS

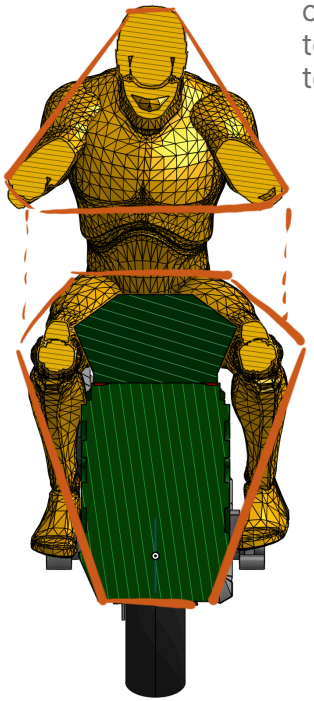


B: Considering the body as an approximate windshield, we can say that it is bowed given the curved shape of the chest, therefore we have **B3.**

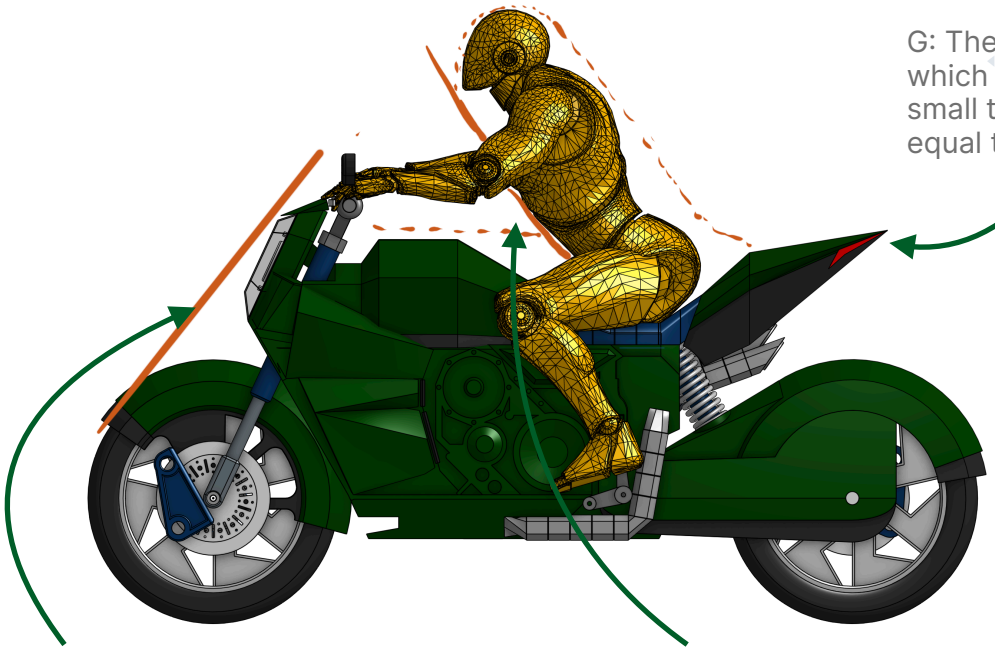
D: The end of bike tapers in very aggressively, similarly to **D1**.

A: The body can be evened out to be **A1** given how thin it is.

C: The body also tapers back giving us **C2.**



H: Looking at the cross-section, we can approximate it to the shape similar to **H4**.



G: The back is very short which equates to a very small trunk space roughly equal to **G5**.

E: The front of the bike is very steep and, drawing an approximate line for it's shape, the closest match is **E1**, as it is very high and squared.

F: Given the position of the user, the windshield peak can be assumed to be **F3** as it has a forward projecting peak-like shape.

$$C_d = 0.16 + 0.0095 \sum_{H}^{I=A} N_i$$

$$C_d = 0.16 + 0.0095(27) = 0.417$$

06 CFD MODEL

Analysis of mesh

Total Cell count: 370,405 (Smaller due to bike dimensions)
Fluid Cells: 370,405
Solid Cells: 15,505
Partial Cells: N/A
Trimmed Cells: 0

Additional physical calculations options

Heat Transfer Analysis: Heat conduction in solids: Off
Flow Type: Laminar and turbulent
Time-Dependent Analysis: Off
Gravity: On
Radiation: Off
Humidity: Off
Default Wall Roughness: 0 micro-metre

Initial conditions:

Static Pressure: 101325 Pa
Temperature: 293.2 K
Velocity: -31.29 m/s in Z direction
Turbulence Intensity: 0.1%
Turbulence Length: 0.007451 m

Engineering goals:

Lift: Force (Y)
Drag: Force (Z)
Vehicle Size: 0.516m^2
Drag Coefficient: (2*Drag Force)/(1.2525*979.06*0.516)*(-1)

Analysis time

Calculation Time: 16:01 mins
Number of Iterations: 151

Results:

Running the CFD model, I achieved the results below. Starting from the **lift** force, the value was only **2.7N** upwards meaning that little downforce acts on the wheels, something that is bad for a car, but excellent for a motorcycle as it helps in keeping the bike balanced when steering. The **drag force** was a bit higher at -**183.7N**, however, what gives us correlation is the **drag coefficient** at **0.594**. While this might be high for a car, it is actually quite reasonable for a motorbike, especially a naked at that.

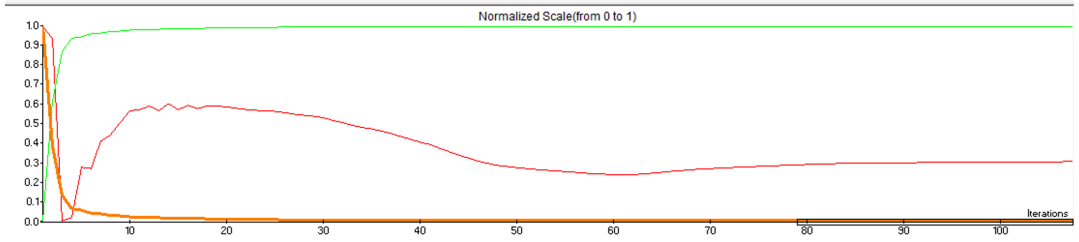


Fig 1: Goals' Value Vs Iteration Plot

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Force (Y) 1	[N]	2.751	2.628	1.793	2.831	100	Yes	1.039	1.155
GG Force (Z) 2	[N]	-183.742	-183.643	-183.795	-183.249	100	Yes	0.546	34.412
Drag Coefficient	[ ]	0.5938039	0.5934832	0.5922115	0.5939752	100	Yes	0.0017637	0.1112110

Fig 2: Table Of Results

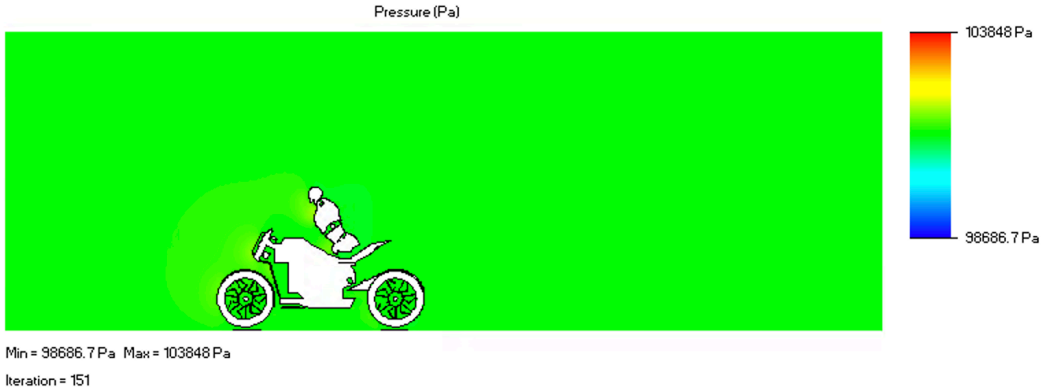
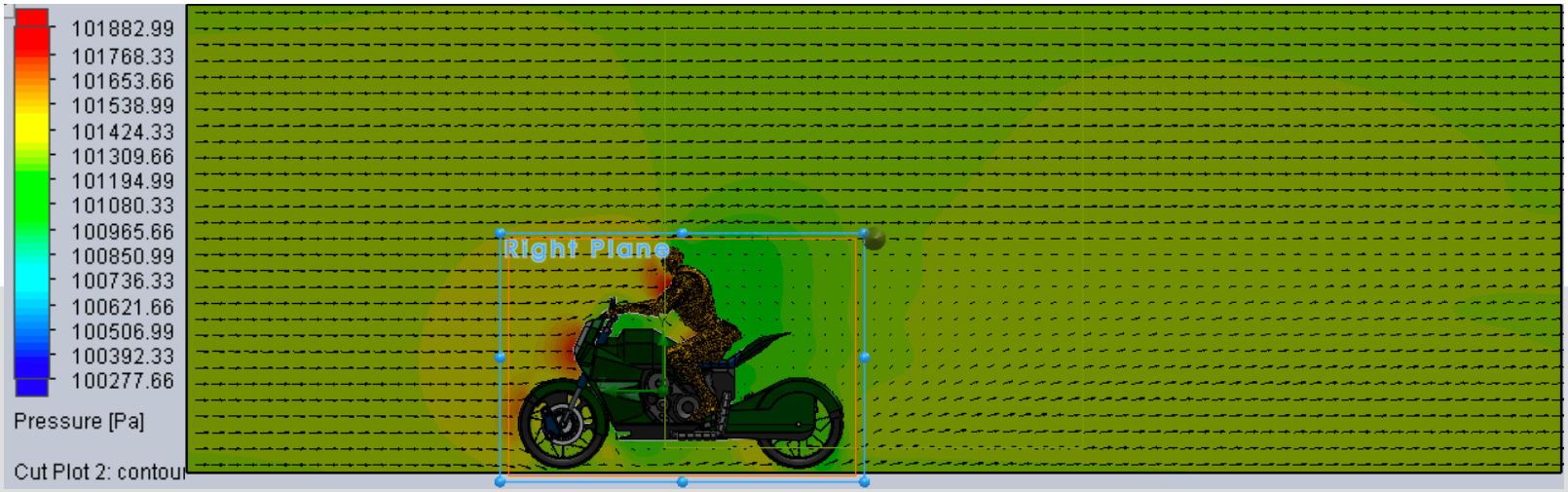
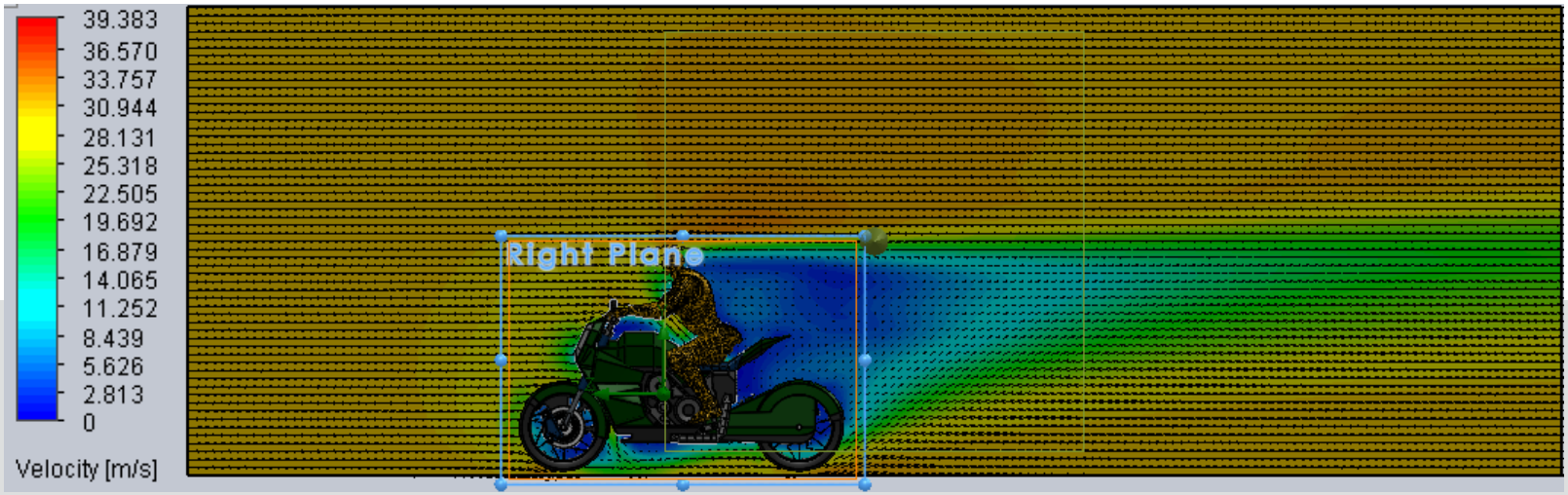


Fig 3: Preliminary Pressure

07 PRESSURE AND VELOCITY PLOT

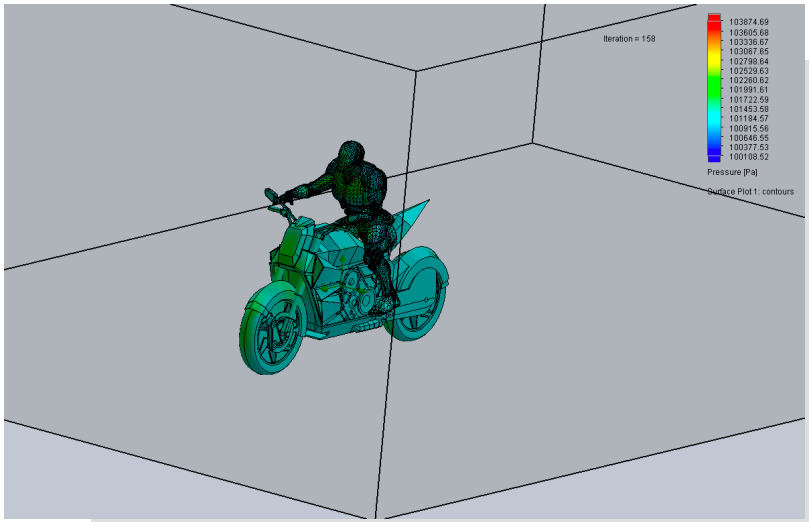


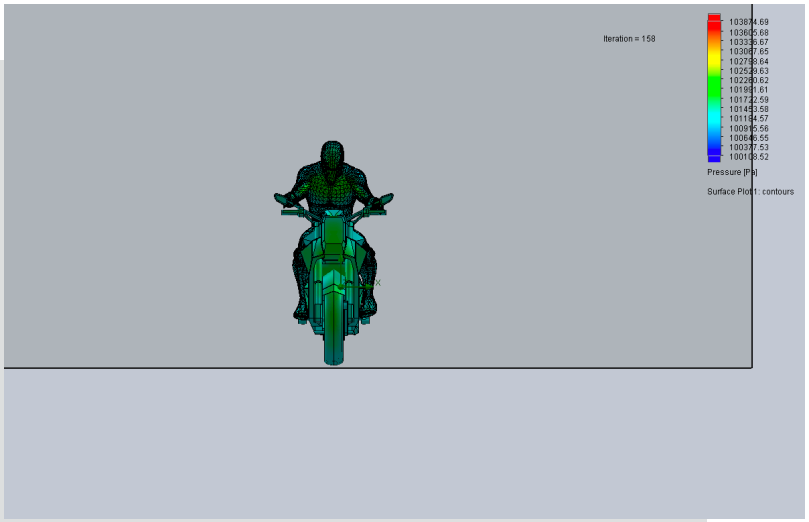
The **Pressure Plot** above shows how **high** pressure zones are formed in 3 key places, in front of the wheel, the headlight, and most importantly, from the user's body. On the other hand, a **large low** pressure zone is formed behind the driver and partially below the bike.



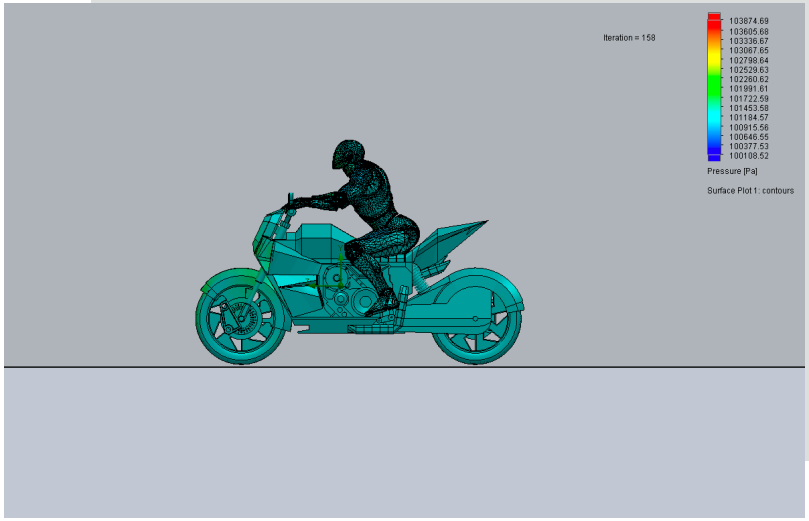
The **Velocity Plot** reveals why the model has such a **low downward force**. The user's body together with the overall shape of the bike has led to the air being dragged **upwards**. The large zone of slow air behind the user is due to a large amount of **turbulence** which results in a lot of drag.

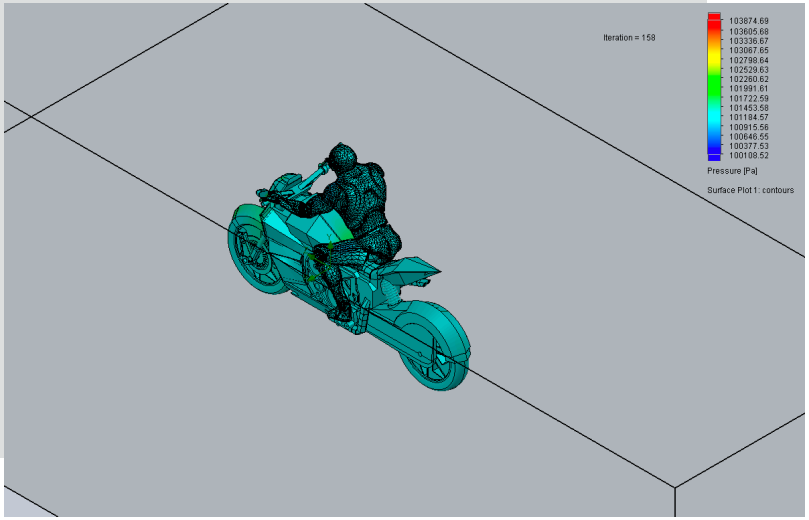
08 3D PRESSURE DISTRIBUTION



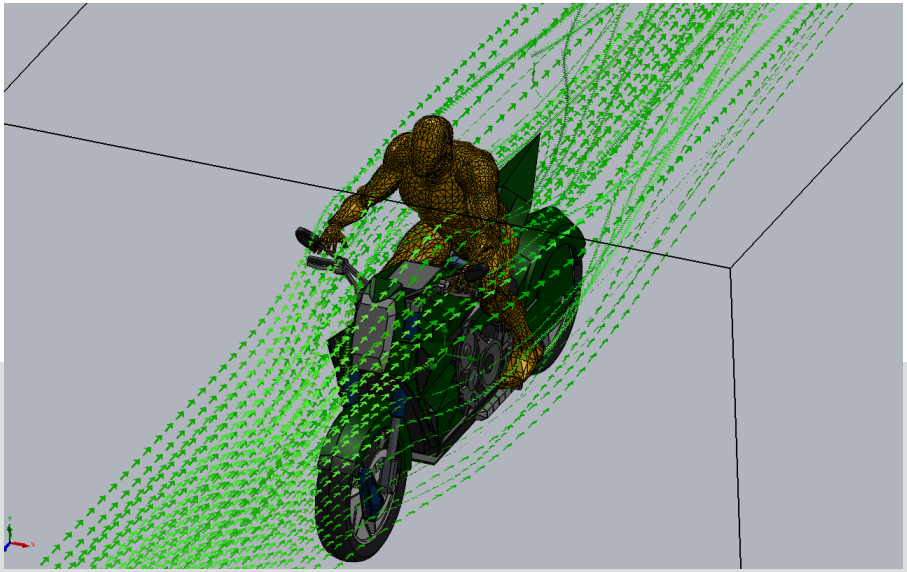


Pressure has been mostly evenly distributed across the bike with some small **higher pressure** areas around the front of the wheel, the headlights and the middle of the chest. However, there are no really high pressure zones mostly due to the fact that a bike is so much **thinner** than a car and so it's easier for the air to flow around it.





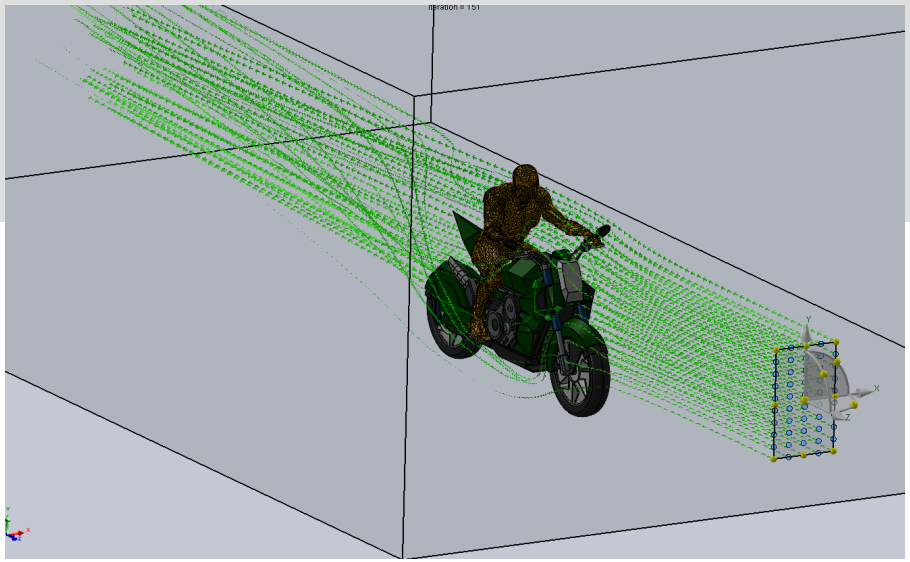
3D FLOW TRAJECTORIES

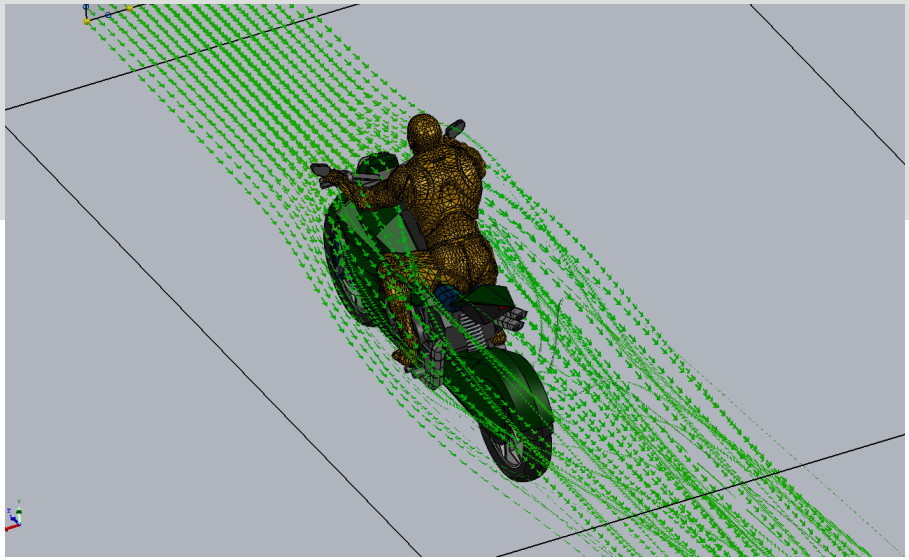


Lots of turbulence

As you can see from looking at the airflow behind the model, the design has resulted in a **very turbulent** airflow being left in its wake. This is largely due to the **user's body** and its position cause air to pocket in both in front and behind him causing plenty of turbulence and thus **drag**.

The low pressure and low velocity zones that you can see behind the body on the **2D plots** have mostly as a result of the turbulent airflow. Another area that has issue with lots of roughness is directly in front of the body between the user and the front light which is a zone where air gets trapped in and circulates onto itself before being able to move around the user's body.





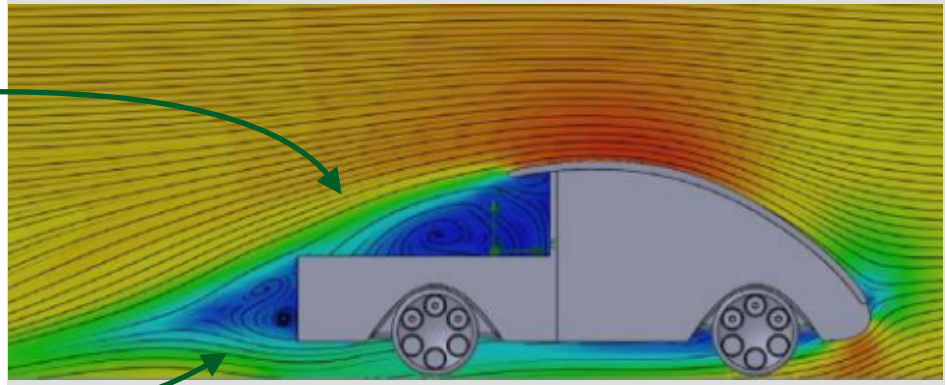
10 COMPARING IT TO REAL WORLD EXAMPLES

What to compare it to?

Motorbikes are by their nature not that aerodynamic. Given all the real world examples were of cars, I knew that I would have to be careful regarding what class I compared it to. If I looked at something like sports cars, I knew that I could not have an **equitable** comparison given the large difference in the range of the drag coefficient. Because of that, I chose to compare it to something like the **Pick-up truck** which has the highest coefficient of drag and so can be the **closest** to my motorcycle.

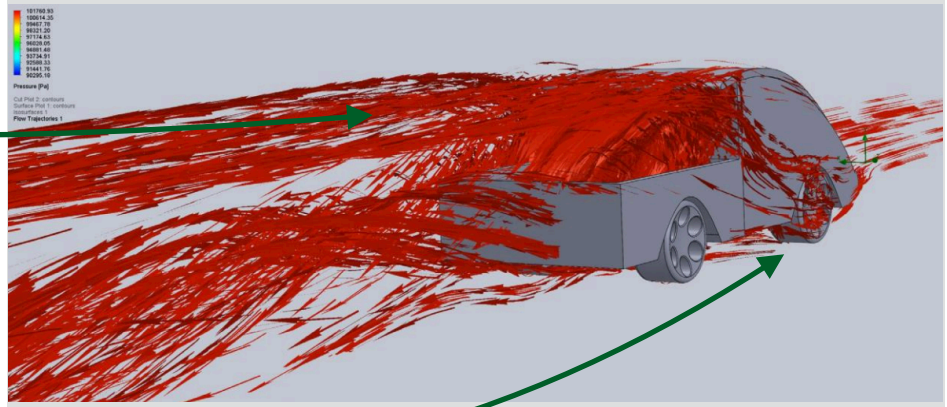
Increases the drag on the vehicle.

Low pressure zone directly behind the top of the roof similarly to how on the motorbike it's behind the person.



This is also due to lots of turbulent air flow behind it.

Airflow leads downwards in the pickup while on the motorbike it's up. This is due to the **lip** behind the roof and overall shape of the body.



Areas where air can get trapped always lead to **more drag** and so should be avoided.

While on the motorbike air gets trapped in-between the front light and the user, on the pick up truck it's **comparatively** getting trapped in both the wheel arches and the trunk.

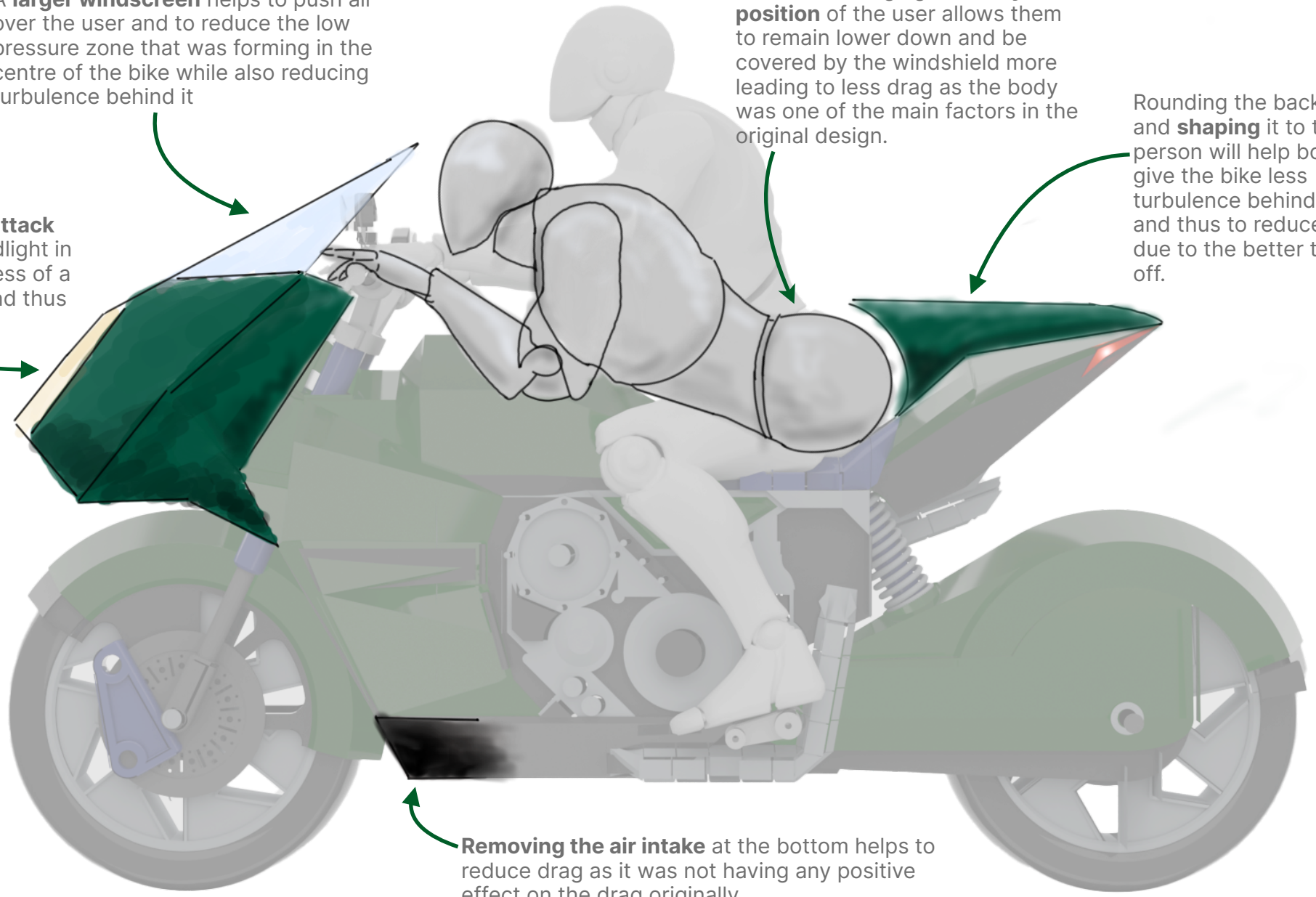
11 IMPROVEMENTS

A **larger windscreen** helps to push air over the user and to reduce the low pressure zone that was forming in the centre of the bike while also reducing turbulence behind it

Moving the seat slightly further back and changing the **body position** of the user allows them to remain lower down and be covered by the windshield more leading to less drag as the body was one of the main factors in the original design.

Rounding the back off and **shaping** it to the person will help both give the bike less turbulence behind it and thus to reduce drag due to the better taper off.

Lower angle of attack on the front headlight in order to create less of a pressure zone and thus reduce drag.



Removing the air intake at the bottom helps to reduce drag as it was not having any positive effect on the drag originally.